

Oscillatory motion is all around us – atoms vibrating in a solid, a bridge swaying in the wind, the motion of pistons of a car and the motion of tides.

This section provides knowledge and understanding of simple harmonic motion, forced oscillations and resonance.

5.3.1 Simple harmonic oscillations

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) displacement, amplitude, period, frequency, angular frequency and phase difference	M4.7
(b) angular frequency ω ; $\omega = \frac{2\pi}{T}$ or $\omega = 2\pi f$	
(c) (i) simple harmonic motion; defining equation $a = -\omega^2 x$ (ii) techniques and procedures used to determine the period/frequency of simple harmonic oscillations	PAG10 e.g. mass on a spring, pendulum
(d) solutions to the equation $a = -\omega^2 x$ e.g. $x = A \cos \omega t$ or $x = A \sin \omega t$	M3.9, M3.12
(e) velocity $v = \pm \omega \sqrt{A^2 - x^2}$ hence $v_{\max} = \omega A$	M2.2
(f) the period of a simple harmonic oscillator is independent of its amplitude (isochronous oscillator)	
(g) graphical methods to relate the changes in displacement, velocity and acceleration during simple harmonic motion.	